Bar-chain models of E_1-chiral quantum groups and the E_N ladder

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1 The Yangian as E_1-chiral quantum group

The RTT Yangian $Y_\hbar(\mathfrak{gl}_2)$ is presented by generators $t_{ij}^{(r)}$, $1 \leq i, j \leq 2$, $r \geq 1$, assembled into the generating series $T(u) = \mathbf{1} + \sum_{r \geq 1} T^{(r)} u^{-r}$ with $T^{(r)} = (t_{ij}^{(r)})$, subject to the RTT relation

$$R(u-v)\, T_1(u)\, T_2(v) = T_2(v)\, T_1(u)\, R(u-v), \qquad R(z) = 1 - \frac{\hbar P}{z}, \tag{1}$$

where $P \in \mathrm{End}(\mathbb{C}^2 \otimes \mathbb{C}^2)$ is the permutation. The special Yangian for $\mathfrak{sl}_2$ is obtained only after imposing the quantum-determinant constraint; this first calculation uses the unreduced RTT algebra. The deformation parameter $\hbar$ is formal, and the specialization $\hbar = 0$ is the commutative degeneration with trivial R-matrix.

Multiplication. The RTT relation (1) defines a $\mathbb{Z}$-filtered associative algebra. For formal $\hbar$ it is not commutative: $[t_{ij}^{(1)}, t_{kl}^{(1)}] = \hbar(\delta_{kj} t_{il}^{(1)} - \delta_{il} t_{kj}^{(1)})$. Multiplication is E_1 and no more.

Coproduct. The spectral coproduct

$$\Delta_z \colon Y_\hbar(\mathfrak{gl}_2) \to Y_\hbar(\mathfrak{gl}_2) \mathbin{\widehat{\otimes}} Y_\hbar(\mathfrak{gl}_2), \qquad \Delta_z(T(u)) = T(u) \mathbin{\dot{\otimes}} T(u-z), \tag{2}$$

that is, $\Delta_z(t_{ij}(u)) = \sum_a t_{ia}(u) \otimes t_{aj}(u-z)$, is coassociative: $(\mathrm{id} \mathbin{\widehat{\otimes}} \Delta_w) \circ \Delta_z = (\Delta_z \mathbin{\widehat{\otimes}} \mathrm{id}) \circ \Delta_{z+w}$. At $z = 0$ it reduces to the RTT Drinfeld coproduct $\Delta_0(T(u)) = T(u) \dot{\otimes} T(u)$.

R-matrix and quasi-triangularity. The Yang R-matrix $R(z) = 1 - \hbar P/z$ satisfies the Yang–Baxter equation

$$R_{12}(z_1 - z_2)\, R_{13}(z_1 - z_3)\, R_{23}(z_2 - z_3) = R_{23}(z_2 - z_3)\, R_{13}(z_1 - z_3)\, R_{12}(z_1 - z_2). \tag{3}$$

It is the fundamental representation-level R-matrix. Algebra-level quasi-triangularity for a module category uses the completed universal R-matrix; on fundamental evaluation modules the displayed finite matrix gives the usual RLL intertwining relation.

Braided module category. For finite-dimensional evaluation modules M_u, N_v of $Y_\hbar(\mathfrak{gl}_2)$, the spectral coproduct (2) defines a fusion product $M \otimes_z N$ by pulling back the $(Y_\hbar \widehat{\otimes} Y_\hbar)$-action. The braiding

$$\beta_{M_u, N_v} \colon M_u \otimes N_v \xrightarrow{R_{M,N}(u-v)\cdot\sigma} N_v \otimes M_u$$

satisfies the braid relation by (3).

What makes $Y_\hbar(\mathfrak{sl}_2)$ genuinely E_1. The RTT relation (1) is *not* derived from a local OPE on a curve. There is no vertex algebra V for which $Y_\hbar(\mathfrak{sl}_2)$ arises as a chiral envelope. The ordering of tensor factors is independent input. In the hierarchy

$$E_1\text{-chiral} \supsetneq E_\infty\text{-chiral (vertex algebras)},$$

the Yangian sits in $E_1 \setminus E_\infty$: genuinely nonlocal.

This example earns the following definition.

2 E_1-chiral quantum groups

Definition 2.1 (Ordered Koszul chiral algebra (Vol I ??, ??)). *An E_1-chiral algebra $\mathcal{A}$ on a smooth curve X is* ordered Koszul *if (i) its ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ over the pure braid Orlik–Solomon operad $\mathrm{OS}(\mathrm{PBr}_n)$ is quadratic-Koszul in the Shelton–Yuzvinsky sense; (ii) the induced A_∞-structure $\{m_k^{\mathrm{ord}}\}_{k\geq 2}$ on $\mathcal{A}$ satisfies the Stasheff boundary-of-associahedra relations; (iii) the R-twisted Σ_n-descent identifying $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ with the symmetric-bar shadow $\overline{B}^{\Sigma}(\mathcal{A})$ holds at the level of chain complexes.*

Proposition 2.2 (Yangian as ordered-Koszul E_1-chiral algebra (Vol I ??, ??)). *On the rational ordered Ran surface of the affine line, the Yangian $Y_\hbar(\mathfrak{g})^{\mathrm{ch}}$ in Drinfeld's second presentation is an ordered Koszul E_1-chiral algebra in the sense of Definition 2.1. The Etingof–Kazhdan avatar is a quantum vertex algebra with S-locality, not an ordinary symmetric chiral algebra. Its braiding operator satisfies*

$$S^{21}(z) = S^{-1}(-z);$$

for the bare rational matrix $R_0(z) = 1 - \hbar P/z$ this requires a scalar normalisation $f(z)$ with $f(z)f(-z) = (1 - \hbar^2/z^2)^{-1}$. Even after this unitarisation, S is nontrivial. Therefore ordinary symmetric descent to an arbitrary smooth curve is not part of the proposition. Global curve versions require separate rational, trigonometric, or elliptic KZ/KZB descent data.

Remark 2.3 (Why the Yangian is the archetypal E_1-chiral input). The five notions of E_1-chiral algebra (strict ChirAss, A_∞ in $\mathrm{End}^{\mathrm{ch}}_{\mathcal{A}}$, Etingof–Kazhdan quantum VA, A_∞ in E_1-chiral, factorization on $\mathrm{Ran}^{\mathrm{ord}}(X)$) coincide on the ordered Koszul locus where the Drinfeld associator acts freely, through Proposition 2.2. The Yangian sits in the complement of E_∞-chiral (standard vertex algebras) and is genuinely nonlocal.

Definition 2.4 (E_1-chiral quantum group). *An E_1-**chiral quantum group** on a smooth curve X is a chiral bialgebra datum $(\mathcal{A}, \mu, \Delta^{\mathrm{ch}}, R, \Phi)$ where:*

(i) *$\mathcal{A}$ is an E_1-chiral algebra on X: an E_1-algebra object in factorisation algebras on X;*

(ii) *μ is the E_1-chiral multiplication (the A_∞-structure $\{m_k\}_{k\geq 2}$ encoding the ordered OPE);*

(iii) *$\Delta^{\mathrm{ch}} : \mathcal{A} \to (\mathcal{A}\widehat{\otimes}\mathcal{A})((z))$ is a completed spectral chiral coproduct on $\mathcal{A}$;*

(iv) *Φ is a fixed Drinfeld associator and Δ^{ch} is coassociative up to conjugation by Φ;*

(v) *$R(z)$ is either the completed universal R-matrix for the chosen module category or its evaluation on specified modules, satisfying the Yang–Baxter and $R\Delta = \Delta^{\mathrm{op}}R$ intertwining identities in that completion.*

An antipode is not part of the chiral bialgebra datum. It is additional Drinfeld-double or Hopf data, and it does not lift automatically to the vertex/chiral level.

The coproduct Δ^{ch} is structure on $\mathcal{A}$. It is not the deconcatenation coproduct on the bar complex $B(\mathcal{A})$: deconcatenation is a structural feature of any bar construction and carries no information about $\mathcal{A}$ beyond what the product already encodes.

Theorem 2.5 (Categorical level shift). *Let $(\mathcal{A}, \mu, \Delta^{\mathrm{ch}}, R, \Phi)$ be an E_1-chiral quantum group datum. Then:*

(a) *$\mathcal{A}$ is an E_1-algebra.*
(b) *On any module category on which Δ^{ch}, R, and Φ converge, $\mathrm{Mod}_{\mathcal{A}}$ is braided monoidal. Its associativity is controlled by Φ, and the spectral dependence comes from the $\mathcal{D}$-module translation parameter.*
(c) *The derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) := \mathrm{ChirHoch}^{\bullet}(\mathcal{A}, \mathcal{A})$ carries the proved E_2-chiral brace action. The two-coloured $\mathsf{SC}^{\mathrm{ch,top}}$ bulk-boundary action on $(Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$ requires the heptagon hypotheses, and a strict E_3^{top} refinement requires explicit topologisation data.*

No single-colour E_3^{top} structure is produced by the definition alone.

Proof. (a) is the definition.

(b) The coproduct Δ^{ch} defines the monoidal product $M \otimes_{\Delta} N$ on $\mathrm{Mod}_{\mathcal{A}}$ by restriction of scalars along Δ^{ch}. Coassociativity up to Φ gives the associator of $\otimes_{\Delta}$. The completed R-matrix and quasi-triangularity give the braiding natural transformation $\beta_{M,N} : M \otimes N \to N \otimes M$ satisfying the hexagon axioms (which follow from the YBE (3) and the pentagon for Φ).

(c) The chiral higher Deligne theorem (Vol II **??**): for an E_1-chiral algebra $\mathcal{A}$ on a smooth curve X, the chiral Hochschild cochains $\mathrm{ChirHoch}^{\bullet}(\mathcal{A}, \mathcal{A})$ carry the canonical E_2-chiral brace structure. The mixed $\mathsf{SC}^{\mathrm{ch,top}}$ universal property is conditional on the two-coloured cobar contracting homotopy, and the strict E_3^{top} action appears only after explicit inner-stress topologisation. The chain-level action is associator-dependent; its cohomological E_2 brace shadow is associator-free. $\square$

3 The bar complex: what it sees and what it does not

Let $\mathcal{A}$ be an E_1-chiral algebra on a smooth curve X.

Proposition 3.1 (Bar complex as E_1 coalgebra [1, Theorem A]). *For an augmented ordered E_1-chiral algebra, the reduced ordered bar coalgebra*

$$\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}}) = \bigoplus_{n\geq 0}(s^{-1}\bar{\mathcal{A}})^{\otimes n}, \qquad d_B = \sum_{k\geq 2}\sum_i m_k^{(i)},$$

$$\Delta(a_1|\cdots|a_n) = \sum_{i=0}^{n}(a_1|\cdots|a_i) \otimes (a_{i+1}|\cdots|a_n).$$

is an E_1 dg coassociative coalgebra on $\mathrm{Ran}(X)$ *at genus* 0. *The differential d_B is a coderivation of* Δ. *Under the usual augmented, conilpotent, completed hypotheses, the cobar construction gives bar-cobar inversion $\Omega\overline{B}^{\mathrm{ord}}(\mathcal{A}) \simeq \mathcal{A}$.*

The two-sided bar model computes $k \otimes^{\mathbf{L}}_{\mathcal{A}} k$ with augmentation boundary conditions. This is distinct from factorisation homology of the line: $\int_{\mathbb{R}} \mathcal{A} \simeq \mathcal{A}$. The identity $\Omega\overline{B}^{\mathrm{ord}}(\mathcal{A}) \simeq \mathcal{A}$ is inversion, not

Koszul duality; the Koszul dual $A^!$ is obtained from bar cohomology together with the Verdier/completed duality data.

Proposition 3.2 (Classical r-matrix from collision residues). *Let $\mathcal{A} = V_k(\mathfrak{g})$ be the Kac–Moody vertex algebra at level k. The genus-0, degree-2 collision residue of the bar differential gives*

$$r(z) = \frac{k\,\Omega}{z}, \tag{4}$$

where $\Omega \in \mathfrak{g} \otimes \mathfrak{g}$ is the Casimir. This $r(z)$ satisfies the classical Yang–Baxter equation.

Proof. The current OPE has two different singular channels:

$$J^a(z)J^b(w) \sim \frac{k\,\delta^{ab}}{(z-w)^2} + \frac{f^{ab}_c J^c(w)}{z-w}.$$

The double pole records the level pairing. The KZ/factorisation connection packages this invariant form as the split Casimir channel $k\,\Omega/(z_1 - z_2)$ on two insertions; this is the normalization used in (4). The CYBE is the codimension-two compatibility on $\mathrm{FM}_3(\mathbb{C})$, equivalently the Jacobi identity for the split Casimir. □

Proposition 3.3 (Curved modular bar curvature). *In the curved modular bar model, under the rank-one clean-intersection or scalar-diagonal hypotheses on the curvature channel, the fibre differential has*

$$d^2_{\mathrm{fib}} = \kappa_{\mathrm{ch}} \cdot \omega_g, \tag{5}$$

where κ_{ch} is the shadow obstruction class and ω_g is the Arakelov $(1,1)$-form on the genus-g surface, normalised so that $\int_{\Sigma_g} \omega_g = 1$. The genus-0 ordered bar differential remains a genuine differential; the curvature belongs to the modular genus-g extension, not to the ordinary dg coalgebra of Proposition 3.1.

The bar complex sees the *product* (fusion, encoded in d_B) and the *classical r-matrix* (from collision residues).

It does *not* see:

(i) the chiral coproduct $\Delta\colon \mathcal{A} \to \mathcal{A} \otimes \mathcal{A}$ (the coproduct encodes splitting, not fusion; it is independent algebraic structure);
(ii) the full quantum R-matrix $R(z)$ (only its classical limit $r(z)$ appears in the shadow tower);
(iii) quasi-triangularity or the antipode.

The shadow obstruction tower $\{F_g\}_{g\geq 1}$, extracted from (5), detects collision data. The coproduct encodes the opposite operation and is invisible to the tower.

Sources of the chiral coproduct: Gaiotto–Rapčák–Zhou [4] construct coproducts on deformed double current algebras of $\mathfrak{gl}_K$ from M2–M5 fusions; Jindal–Kaubrys–Latyntsev [6] construct vertex coproducts on CoHAs of quivers with potential, recovering Drinfeld's deformed coproduct for ADE quivers.

4 The E_N ladder

For an E_1-chiral algebra $\mathcal{A}$ on X, the following structures form a ladder only after the relevant direction has been made topological. The mixed operad $\mathsf{SC}^{\mathrm{ch,top}}$ is two-coloured and directional; Dunn additivity enters only after explicit topologisation data have replaced the chiral direction by a topological one.

E_1: the boundary algebra

$\mathcal{A}$ is E_1-chiral: an E_1-algebra in factorisation algebras on X. The bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ is the E_1 dg coassociative coalgebra of Proposition 3.1.

$E_1 \to E_2$: Hochschild and Deligne

Theorem 4.1 (E_2 on the derived centre)**.** *The chiral derived centre*

$$Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = C^{\bullet}_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$$

carries the E_2-chiral brace structure. This is the chiral Deligne–Tamarkin output, after the chosen associator has strictified the brace action:

(a) *the holomorphic part: the chiral algebra structure of $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ on X;*
(b) *the ordered part: the A_∞-algebra structure in the transverse direction.*

The pair $(Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}),\ \mathcal{A})$ is the Swiss-cheese datum: the bulk acts on the boundary through the two-coloured chiral-topological structure.

$E_2 \to E_3$-chiral: chiral higher Deligne

The E_3-chiral structure on $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = \mathrm{ChirHoch}^{\bullet}(\mathcal{A}, \mathcal{A})$ is governed by the chiral higher Deligne theorem (Vol II **??**). Its proved core is the E_2-chiral brace action. The $\mathsf{SC}^{\mathrm{ch,top}}$ universal property and any strict E_3^{top} refinement require the heptagon and topologisation hypotheses stated in Vol II **??**.

Theorem 4.2 (Chiral higher Deligne (Vol II **??**))**.** *Let $\mathcal{A}$ be an E_1-chiral algebra on a smooth complex curve X in the Koszul locus, and fix a Drinfeld associator. Then $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ carries the proved E_2-chiral brace action. If the two-coloured $(\mathsf{SC}^{\mathrm{ch,top}})^!$ cobar contracting homotopy exists, the pair $(Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$ satisfies the $\mathsf{SC}^{\mathrm{ch,top}}$ universal bulk-boundary property. If, in addition, inner-stress topologisation data are supplied, this mixed datum refines to a strict E_3^{top} structure.*

Theorem 4.3 (E_3-chiral realisation for Kac–Moody [8])**.** *For $\mathcal{A} = V_k(\mathfrak{g})$ at non-critical level $k \neq -h^\vee$, holomorphic Chern–Simons theory on $X \times \mathbb{R}$ provides the 3d HT bulk. The factorisation algebra of observables on $X \times \mathbb{R}$ gives the mixed 3d HT bulk, and its restriction to the boundary $X \times \{0\}$ recovers $V_k(\mathfrak{g})$. This geometric realisation agrees with the chiral higher Deligne mixed structure of Theorem 4.2 on cohomology; the chain-level comparison is associator-dependent and model-specific.*

Kac–Moody has a proved gauge-theoretic realisation at non-critical level. Drinfeld–Sokolov and freely generated PVA cases require their own BRST or Poisson-sigma hypotheses in addition to Dunn additivity.

E_3-chiral $\to$ E_3-topological: topologisation

Theorem 4.4 (Topologisation for Kac–Moody)**.** *Let $\mathcal{A} = V_k(\mathfrak{g})$ at non-critical level $k \neq -h^\vee$. Write $T(z) = \{Q, G(z)\}$ where $T(z)$ is the Sugawara conformal vector, Q is the BRST operator, and $G(z)$ is the antighost. Then, on the topologised cohomology model:*

(a) *Translations along X are Q-exact on the chain-level antighost refinement of Vol I ??, so the complex structure of X is irrelevant in the chain-level BRST cohomology.*

(b) *The mixed structure on $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ (Theorem 4.2, Theorem 4.3) descends to an E_3-topological structure after the chiral translation has become Q-exact.*

(c) *The resulting E_3-topological theory is Chern–Simons at level $k + h^\vee$.*

Theorem 4.5 (Topologisation tower with hypotheses (Vol I ??, Vol II ??))**.** *On a chain model carrying the required stress-tower data:*

(i) *class G (Heisenberg, nonzero level) reaches E_3^{top} through the abelian Sugawara field;*

(ii) *class L ($V_k(\mathfrak{g})$, $k \neq -h^\vee$) E_3^{top} on the topologised cohomology model; a strict original-chain lift requires the antighost homotopy equation;*

(iii) *class C ($\beta\gamma$, bc) E_3^{top} via FMS bosonisation reducing to class G.*

In the weight-completed category:

(iv) *class M (Virasoro, $\mathcal{W}_N$) E_3^{top} via half-BRST and the MC5 weight-completed mechanism (?? + ??).*

At critical level $k = -h^\vee$, Sugawara degenerates and the derived centre retains its E_2-chiral brace shadow. The Feigin–Frenkel centre supplies a commutative horizontal sector, but the non-critical topologisation ladder is absent: no E_3^{top} rung is attained for the full affine algebra without a non-critical conformal vector. Higher-spin extension (Vol II ??): the first k pairwise commuting inner stress tensors, with coherent homotopies and continuous action on the chosen chain model, yield E_{k+2}^{top}. For $\mathcal{W}_\infty$ this is a weight-filtered inverse-limit statement under the same convergence hypotheses. The raw original complex requires a separate convergence theorem.

Conjecture 4.6 (Topologisation for general conformal chiral algebras, original complex class M)**.** *Let $\mathcal{A}$ be a class-M chiral algebra at non-critical level. The weight-completed E_3^{top} structure of Theorem 4.5(iv) lifts to an E_3^{top} structure on the original complex. For Virasoro and principal $\mathcal{W}_N$ at generic central charge, this original-complex lift remains conjectural. BP (minimal $\mathcal{W}^{(2)}(\mathfrak{sl}_3)$) has a strict chain-level original-complex proof via branched-cover integralisation plus Galois descent (Vol II ??), indicating the route for general class M is branched-cover reduction to class L upstairs.*

At the *critical level* $k = -h^\vee$, the Sugawara construction degenerates. No non-critical conformal vector is available, so the non-critical topologisation ladder fails. The derived centre remains E_2-chiral and has the enlarged Feigin–Frenkel horizontal centre; this commutative sector is a degenerate critical-level topological shadow, not an E_3^{top} structure on the full affine algebra.

Remark 4.7 (The generic case is $\mathrm{SC}^{\mathrm{ch,top}}$, not E_3). E_3-topological requires *both* a conformal vector at non-critical level *and* a 3d holomorphic-topological bulk theory. Chiral algebras lacking these inputs, such as critical-level Kac–Moody $V_{-h^\vee}(\mathfrak{g})$, E_1-chiral algebras (Yangians), and Calabi–Yau functor outputs, carry the generic first-class structure $\mathrm{SC}^{\mathrm{ch,top}}$, in which the holomorphic and topological colours remain distinct.

The Heisenberg algebra $\mathcal{H}_k$ ($k \neq 0$) and lattice VOAs are *not* stuck at $\mathrm{SC}^{\mathrm{ch,top}}$: $\mathcal{H}_k$ carries the abelian Sugawara conformal vector $T = \frac{1}{2k}{:}JJ{:}$ ($c = 1$) and abelian holomorphic Chern–Simons theory provides the 3d HT bulk, so $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{H}_k)$ reaches E_3-topological (the same proof as for non-abelian Kac–Moody, Theorem 4.4).

5 The modular operad and the curved-Dunn range

The chiral-A_∞ modular operad is the ambient object for the $E_2 \to E_3$ step. Curved Dunn additivity has a conditional higher-genus vanishing range and a separate genus-one curved comparison; these are not the same assertion.

Theorem 5.1 (Chiral-A_∞ modular operad (Vol II `ch:modular-swiss-cheese-operad`))**.** *The modular operad $\mathcal{O}^{\mathrm{ch}\text{-}A_\infty}$ exists and its algebras consist of:*

(a) *an E_1-chiral algebra structure on X (governed by $\mathsf{SC}^{\mathrm{ch,top}}$ on within-surface strata);*
(b) *an A_∞-algebra structure in the transverse direction $\mathbb{R}$;*
(c) *compatibility between transverse A_∞-operations and the E_1 ordering on X;*
(d) *at genus $g \geq 1$: curvature $d^2_{\mathrm{fib}} = \kappa_{\mathrm{ch}}(\mathcal{A}) \cdot \omega_g$ as the bar intrinsic obstruction to extending the genus-0 structure to higher genera.*

The genus-0 part of $\mathcal{O}^{\mathrm{ch}\text{-}A_\infty}$ is $\mathsf{SC}^{\mathrm{ch,top}} \times E_1^{\mathrm{tr}}$.

Theorem 5.2 (Conditional curved-Dunn higher-genus range)**.** *Assume the modular-bootstrap degree-two vanishing and the low-degree bridge comparison between the modular-bootstrap complex and the curved-Dunn twisting-cochain complex. At genus $g \geq 2$, the curved modular bar complex of $\mathcal{O}^{\mathrm{ch}\text{-}A_\infty}$ factorises as a twisted Eilenberg–Zilber tensor product*

$$\mathbf{B}_{\mathrm{mod}}(\mathcal{O}^{\mathrm{ch}\text{-}A_\infty}) \;\simeq\; \mathbf{B}_{\mathrm{mod}}(\mathsf{SC}^{\mathrm{ch,top}}) \;\otimes^{\tau}\; \mathbf{B}(E_1^{\mathrm{tr}}),$$

with twisting cochain τ built recursively from R-matrix monodromy at each genus-raising clutching. In this range the curved-Dunn obstruction $H^2(\mathrm{TCo}^{\bullet}_{g}(\mathcal{A}), d_0) = 0$ transports from the modular-bootstrap complex via the explicit bridge map of Vol II **??***; the vanishing is therefore conditional on that bridge. The genus-one statement is the separate curved tensor-product comparison of Vol II* **??***; the uniform H^2-vanishing statement requires the higher-genus bridge.*

The unified chiral quantum group

The following families supply chiral bialgebra or line-algebra data (Yangian $Y_\hbar(\mathfrak{g})$, affine Yangian $Y_\hbar(\hat{\mathfrak{g}})$, shifted Yangian $Y_\mu(\mathfrak{g})$, truncated shifted $Y^\lambda_\mu(\mathfrak{g})$ = BFN Coulomb branch, finite W-algebra $W^{\mathrm{fin}}(\mathfrak{g}, f)$ on the topological line, principal affine $W_k(\mathfrak{g}, f_{\mathrm{prin}})$, non-principal affine $W_k(\mathfrak{g}, f)$, Bershadsky–Polyakov $W_k(\mathfrak{sl}_3, f_{\mathrm{min}})$, orthogonal coideal $\mathcal{Q}^\theta$). They are organized by the four-parameter notation $\mathcal{Q}^{k,f,\mu}_{\mathfrak{g}}$ when the ambient category is fixed. The finite W-algebra row is topological E_1 data until a chiral BRST lift is chosen.

Theorem 5.3 (Chiral bialgebra comparison, ordered Koszul locus (Vol I **??**))**.** *On the ordered Koszul locus (Definition 2.1), the three structures: vertex R-matrix data, A_∞ chiral operations, and a completed spectral chiral coproduct $\Delta^{\mathrm{ch}} : \mathcal{A} \to (\mathcal{A} \widehat{\otimes} \mathcal{A})((z))$, are linked by comparison maps after fixing a Drinfeld associator, completion, spectral coordinate, and bialgebra transport datum. Any "two determine the* *third" statement means that the corresponding Maurer–Cartan transport problem has been solved in that completed model. The* GRT_1 *torsor changes associator/formality data; it does not create the spectral parameter or turn bar deconcatenation into the Yangian coproduct. A Hopf antipode is additional structure and is obstructed at the vertex-algebraic level in the* $\mathcal{W}_{1+\infty}$ *family.*

Theorem 5.4 ($\mathfrak{gl}_N$ chiral bialgebra, conditional range (Vol I **??**)). *The principal* $\mathcal{W}_N$ *is reconstructed as the intersection of Feigin–Frenkel screening kernels inside a rank-*$(N-1)$ *Heisenberg,*

$$\mathcal{W}_N \;=\; \bigcap_{i=1}^{N-1} \ker Q_{\alpha_i} \;\subset\; \Pi_{\mathrm{FF}},$$

after the central Heisenberg convention has been fixed and removed. Equivalently, in the rectangular Y*-algebra notation it is the* $(K, L) = (1, N)$ *fibre. The chiral coproduct is obtained by transport from the Gaiotto–Rapčák–Zhou/Yangian side only under the screening compatibility, coideal/truncation stability, and explicit transport map* θ_N*; it is not obtained by simply restricting a primitive coproduct on the screening parent. The datum is not a chiral Hopf algebra; the antipode lift is obstructed. Non-principal* $\mathcal{W}$*-hook types require parabolic-screening input.*

Theorem 5.5 (Class-M weight-completed chiral bialgebra (Vol I **??**)). *For* $\mathcal{W}_{1+\infty}$ *at generic nondegenerate parameters, the R-matrix, chiral* A_∞ *operations, and spectral chiral coproduct form a chiral bialgebra datum in the weight-completed category. The direct-sum class-M chain-level lift is obstructed at weight* 4 *by nonvanishing* $S_4(\mathrm{Vir}_c) = 10/[c(5c+22)]$. *Degenerate or truncation loci require separate analysis.*

Theorem 5.6 (Spectral–categorical bridge (Vol I **??**)). *The* E_2*-Gerstenhaber braiding on* $\mathrm{ChirHoch}^\bullet(\mathcal{A}, \mathcal{A})$ *from chiral higher Deligne is associator-dependent: trivial on* H^0*, homotopic on cochains. The categorical braiding is recovered as the* $\hbar$*-expansion of the* R*-matrix braiding under the* GRT_1 *torsor action of Drinfeld associators. Position-dependence on* $\mathrm{Mod}_{\mathcal{A}}$ *arises from the* $\mathcal{D}$*-module structure via the KZB realisation of the spectral parameter.*

Theorem 5.7 (Class-M global triangle, weight-completed (Vol II **??**)). *For class-M* $\mathcal{A}$ *at generic central charge,*

$$\mathcal{A}_{\mathrm{bulk}} \;\simeq\; Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}_{\mathrm{boundary}}) \;\simeq\; \mathrm{HH}^*_{\mathrm{ch}}(\mathcal{A}^!_{\mathrm{line}})$$

holds in the weight-completed category (Vol I **??** *+* **??***). At chain level on the original complex, the DS–Hochschild bridge (Vol II* **??***) reduces the class-M identification to the class-L identification under its SDR and mixed-action hypotheses for principal and hook-type Drinfeld–Sokolov reductions (Virasoro, principal* $\mathcal{W}_N$*, hook-type* $\mathcal{W}_k(\mathfrak{g}, f)$*).*

6 The comparison triangle for $\mathcal{W}_{1+\infty}$

The bar complex detects multiplication and collision residues. It does not manufacture the chiral coproduct. For $\mathcal{W}_{1+\infty}$ the stable comparison has three independent inputs:

$$\begin{tikzcd} & A_\infty\text{-operations } \{m_k\} & \\ R\text{-matrix } R(z) \arrow[ur] \arrow[rr] & & \text{spectral coproduct } \Delta_z. \end{tikzcd} \tag{6}$$

The arrows in (6) are comparison maps after choices of associator, completion, coordinates, and normalisation. They are not canonical inverse functors on an undeclared category.

The R-matrix input and the A_∞ obligation

The structure function of $\mathcal{W}_{1+\infty} \simeq Y(\widehat{\mathfrak{gl}}_1)$ is

$$g(u) = \frac{(u-h_1)(u-h_2)(u-h_3)}{(u+h_1)(u+h_2)(u+h_3)}, \qquad h_1+h_2+h_3 = 0.$$

Set

$$\sigma_2 = h_1h_2 + h_1h_3 + h_2h_3, \qquad \sigma_3 = h_1h_2h_3, \qquad \psi(u) = 1 + \sigma_3 \sum_{j\geq 0} \psi_j u^{-j-1}.$$

The triality parameters $\lambda_i = -\psi_0\sigma_3/h_i$ satisfy $\lambda_1^{-1}+\lambda_2^{-1}+\lambda_3^{-1} = 0$, and the central charge convention is

$$c = -\sigma_2\psi_0 - \sigma_3^2\psi_0^3 = 1 + \prod_{i=1}^{3}(\lambda_i - 1).$$

The MO R-matrix on the charge-n Fock space $\mathcal{F}_n$ has diagonal eigenvalues $R_{\lambda\lambda}(u) = \prod_{s\in\lambda} g(u+c(s))$ where $c(s) = h_1 i + h_2 j$ is the content of box $s=(i,j)$ in partition λ [3].

A chain-level reconstruction of the operations m_n from this R-matrix requires a top-degree form on the Stasheff cell K_n, regularisation along collision faces, and Stokes boundary identities whose boundary terms are the Stasheff relations. The schematic product $\bigwedge_{i<j} d\log(z_i - z_j)$ is not such a form: it has degree $n(n-1)/2$, while $\dim K_n = n-2$. Thus the R-matrix-to-A_∞ arrow in (6) is a proof obligation until the degree-matched propagator and its boundary calculus are supplied.

The bar dual gives multiplication

Proposition 6.1 (Deconcatenation and the dual product)**.** *The ordered bar complex*

$$B^{\mathrm{ord}}(\mathcal{W}_{1+\infty}) \;=\; T^c(s^{-1}\overline{\mathcal{W}_{1+\infty}})$$

with bar differential $d_B = \sum_{k\geq 2} m_k^{(i)}$ and deconcatenation coproduct has a completed linear dual

$$\big(B^{\mathrm{ord}}(\mathcal{W}_{1+\infty})\big)^\vee.$$

Deconcatenation dualises to the convolution product on this dual. It does not produce the spectral coproduct on the affine Yangian.

Proof. For a coalgebra C, the dual $C^\vee$ carries multiplication $(f * g)(c) = (f\otimes g)(\Delta_C c)$. On the completed ordered bar dual this reads

$$(f*g)([v_1|\cdots|v_n]) = \sum_{i=0}^{n} (-1)^{|g|(|v_1|+\cdots+|v_i|)} f([v_1|\cdots|v_i])\, g([v_{i+1}|\cdots|v_n]).$$

It is a convolution product, not a spectral coproduct. A coproduct on a Koszul dual algebra requires additional bialgebra or factorisation data. The line Koszul dual

$$\mathcal{A}^{!}_{\mathrm{line}} \;=\; H^{*}\big(B^{\mathrm{ord}}(\mathcal{W}_{1+\infty})\big)^{\vee}$$

is therefore not identified with the original boundary algebra by bar-cobar inversion. The five objects $\mathcal{A}$, $B(\mathcal{A})$, $H^{*}B(\mathcal{A})$, $\mathcal{A}^{!}$, and $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ remain distinct. □

The affine-Yangian coproduct input

Proposition 6.2 (Stable-envelope spectral coproduct)**.** *Let $\mathcal{F}_a$ be the completed equivariant K-theoretic Fock module on which the Maulik–Okounkov stable-envelope R-matrix $R^{\mathrm{MO}}_{a,b}(z)$ acts. After a chamber and normalization are fixed, the Gauss factors of $R^{\mathrm{MO}}_{a,b}(z)$ give the Schiffmann–Vasserot spectral coproduct*

$$\Delta^{\mathrm{SV}}_{z} : Y(\widehat{\mathfrak{gl}}_1) \;\longrightarrow\; Y(\widehat{\mathfrak{gl}}_1)\,\widehat{\otimes}\,Y(\widehat{\mathfrak{gl}}_1) \tag{7}$$

on the completed affine Yangian. In Drinfeld-current notation, the current formulas are not naive all-mode identities. They are normal-ordered shorthand after completion, chamber choice, and stable-envelope scalar normalisation. In the Heisenberg-corrected low-mode sector one has

$$\Delta(\psi_3) = \Delta_0(\psi_3) + 6\sigma_3 \sum_{m>0} m\, J_m \otimes J_{-m},$$

where Δ_0 denotes the tensor-shifted primitive part. The remaining Drinfeld-current formulas are read in this same completed normal-ordered sense.

Proof. This imported geometric input comes from stable envelopes. Deconcatenation in the bar construction does not supply it. The stable-envelope R-matrix satisfies the Yang–Baxter equation and intertwines the two tensor-order actions. Its Gauss decomposition, in the normalization used by Schiffmann and Vasserot, identifies the diagonal and triangular factors with completed Drinfeld currents of $Y(\widehat{\mathfrak{gl}}_1)$. The Heisenberg correction displayed above is the first low-mode place where the completed coproduct differs from a raw current substitution. Coassociativity and quasi-triangularity are part of the stable-envelope tensor-functor formalism after completion. □

Remark 6.3 (What the bar comparison must prove). The comparison triangle (6) leaves (7) as external input. It asks for a chain-level lift of the already constructed spectral coproduct to the chosen chiral model. For class L, the ordinary bar-cobar comparison is a quasi-isomorphism under the usual conilpotence and completion hypotheses, so the lift is available on the stated chain model. For class M, the direct-sum complex is not the right ambient category. The nonzero shadow coefficient

$$S_4(\mathrm{Vir}_c) \;=\; \frac{10}{c(5c+22)}$$

is the first visible obstruction to treating the direct-sum bar-cobar map as a chain-level equivalence. The correct comparison uses the weight-completed or pro-completed model.

Theorem 6.4 (Weight-completed class-M bialgebra scope)**.** *For $\mathcal{W}_{1+\infty}$ at the generic nondegenerate levels covered by Vol I* **??***, the completed Schiffmann–Vasserot coproduct* (7) *lifts to the weight-completed*

A_∞ model used in that theorem. The resulting object is a completed chiral bialgebra with associator and R-matrix. The raw direct-sum complex is outside this theorem.

Proof. Vol I **??** supplies the completed identification between the $\mathcal{W}_{1+\infty}$ side and the affine-Yangian side. Proposition 6.2 supplies the spectral coproduct on the Yangian side. Transporting it across that completed comparison gives the stated completed chiral bialgebra structure. No assertion is made for the direct-sum chain complex, where the weight-four shadow obstruction remains visible. □

Remark 6.5 (Positive half and full affine Yangian). The Jindal–Kaubrys–Latyntsev vertex coproduct gives a chain-level construction for the positive half $Y^+(\widehat{\mathfrak{gl}}_1) \simeq \mathrm{CoHA}(\mathbb{C}^3)$. Passing from the positive half to the full affine Yangian requires the compatible opposite half, the Drinfeld double pairing, and the completed topology. The positive-half construction alone supplies only the positive-half coproduct.

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